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# Stereodivergent access to non-natural α-amino acids via enantio- and *Z*/*E*-selective catalysis

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The precise control of *Z* and *E* configurations of the carbon-carbon double bond in alkene synthesis has long been a fundamental challenge in synthetic chemistry, even more pronounced when simultaneously striving to achieve enantioselectivity [(*Z*,*R*), (*Z*,*S*), (*E*,*R*), (*E*,*S*)]. Moreover, enantiopure non-natural α-amino acids are highly sought after in organic and medicinal chemistry. In this study, we report a ligand-controlled stereodivergent synthesis of non-natural α-quaternary amino acids bearing trisubstituted alkene moieties in high yields with excellent enantioselectivity and *Z*/*E* selectivities. This success is achieved through a palladium/copper–cocatalyzed three-component assembly of readily available aryl iodides, allenes, and aldimine esters by simply tuning the chiral ligands of the palladium and copper catalysts.

Twenty proteinogenic amino acids pave the foundation of the origin and evolution of life. Thus, entries to optically active non-natural α-amino acids have numerous applications to the study on the pivotal role of amino acids in the life sciences, drug development, protein engineering, and biomaterial research (*1*–*9*). In this domain, the incorporation of latent (bio)orthogonal reactive alkene functional groups into α-amino acids not only enriches their biological properties but also establishes a versatile platform that enables multiple transformations for generating structurally diverse non-natural α-amino acids (Fig. 1A) (*10*–*14*). It is known that molecules with different stereochemistry (*Z* and *E*, *R* and *S*) exhibit distinctly different biological activities (*15*, *16*). Therefore, the development of an efficient and general catalytic strategy for the stereodivergent synthesis of chiral non-natural α-amino acids bearing stereodefined alkene moieties [(*Z*,*R*), (*Z*,*S*), (*E*,*R*), (*E*,*S*)] is highly desirable. In asymmetric catalysis, chiral ligands have been widely utilized in creating different types of chiral entities. However, achieving full stereocontrol over both the central chirality and C=C bond configuration continues to pose a challenge.

The precise control of both *Z* and *E* configurations of C=C bonds has long been a persistent challenge in organic chemistry because of thermodynamic and kinetic issues (*17*, *18*). Among the limited methods that exist (*19*–*24*), chemistry involving π-allyl metal intermediates may, in principle, offer a direct and elegant strategy for the selective preparation of *Z*- or *E*-olefinic compounds bearing chiral centers (Fig. 1B) (*25*–*37*). The configuration of the resulting alkenes is primarily determined by the selective formation of *syn*- or *anti*-π-allyl metal intermediates, as well as by the selective capture of these intermediates by prochiral nucleophiles. For 1-monosubstituted π-allyl-metal intermediates, the relative thermodynamic stabilities of the intermediates contribute to the dominance of *syn*-π-allyl-metal intermediates, and chiral *E*-disubstituted olefinic compounds [(*E*,*R*) or (*E*,*S*)] are the predominant products (Fig. 1B, left) (*25*–*35*). However, capturing the thermodynamically unstable *anti*-π-allyl-metal intermediates with prochiral nucleophiles for the generation of chiral *Z*-disubstituted olefinic compounds [(*Z*,*R*) or (*Z*,*S*)] presents a great challenge owing to unfavorable R/[TM/L*] strain, and advancements in this area have been lagging (R, alkyl or aryl group; TM, transition metal; L*, chiral ligand) (Fig. 1B, right). Although transition metal–catalyzed *Z*-retentive allylic substitution reactions have been realized (*36*, *37*), asymmetric versions have only recently been reported by You's group (*38*, *39*). By contrast, for 1,2-disubstituted π-allyl-metal intermediates, R/[TM/L*] strain and 1,2-steric repulsion are present in the *anti*- and *syn*-intermediates, respectively (Fig. 1C). As a result, the *anti*- and *syn*-π-allyl-metal intermediates are often energetically close, adding extra complexities to the control of *Z* or *E* selectivity in the to-be-formed trisubstituted alkenes (*40*, *41*). To date, the divergent synthesis of both *Z*- and *E*-alkenes through the selective formation and capture of *anti*- or *syn*-π-allyl metal intermediates remains an elusive goal. Furthermore, this task becomes more challenging for the full stereodivergent synthesis of all four stereoisomers of enantiopure *Z*- and *E*-trisubstituted alkenes [(*Z*,*R*), (*Z*,*S*), (*E*,*R*), (*E*,*S*)] from just a single set of starting materials.

Allenes manifest distinctive properties and reactivities because of their cumulated diene system (*42*–*44*). In this regard, a palladium-catalyzed carbopalladation process of allenes with aryl iodides represents a direct approach to generate the 1,2-disubstituted *syn*- or *anti*-π-allyl-palladium intermediates, albeit with limited control over their distributions and reactivities (*45*–*47*). On the basis of our efforts in allene chemistry and bimetallic catalysis (*48*–*51*), we envisaged that the thermodynamic stabilities of the *syn*- and *anti*-π-allyl-palladium intermediates and their reactivities toward prochiral nucleophiles could be comprehensively regulated by using two chiral catalysts. Here, we report a Pd/Cu synergistic catalytic system that enables the stereodivergent synthesis of enantiopure non-natural α-quaternary amino acids bearing *Z*- and *E*-trisubstituted alkene moieties through a three-component assembly with aryl iodides, simple allenes, and α-amino acid–derived aldimine esters (Fig. 1D). Two chiral catalysts could exert comprehensive stereocontrol over the configurations of the C=C bond and chiral center. This modular catalytic protocol allows access to all four stereoisomers of non-natural α-quaternary amino acids bearing trisubstituted alkene moieties in high yields with excellent enantioselectivities and *Z*/*E* selectivities from the same set of readily available starting materials with judicious ligand combinations.

## Substrate scope exploration

At the beginning of this investigation, we used iodobenzene **1a**, 4-methylphenylpropadiene **2a**, and aldimine ester **3a** as model substrates, with $Cs_2CO_3$ as the base and 1,2-dichloroethane as the solvent under a Pd/Cu dual catalytic system for this three-component assembly process (Fig. 2 and table S1). The chiral palladium catalyst was used to sequentially activate iodobenzene **1a** and allene **2a** (*43*–*45*), and the chiral copper catalyst was specifically used for the activation of aldimine ester **3a**. After a comprehensive evaluation of the two chiral catalysts, two sets of Pd/Cu synergistic catalytic systems were established (tables S1 to S3). The enantiopure non-natural α-quaternary amino acid (*Z*)-**4a** bearing a *Z*-trisubstituted alkene moiety was successfully obtained in 92% yield, 97:3 *Z*/*E* selectivity and 99% enantiomeric excess (e.e.) by using the combination of Cu/($R$,$R_p$-**L1**) and Pd/(*R*)-**L2**. The corresponding *E*-trisubstituted analog (*E*)-**4a** was also prepared in 89% yield, 98:2 *E*/*Z* selectivity, and 98% e.e. by combining Cu/($R$,$R_p$-**L1**) and Pd/($S_p$,*R*)-**L3** (Fig. 2). The substrate compatibility was then investigated under two optimal reaction conditions in a stereodivergent manner. First, the substrate scope of iodobenzene **1** was evaluated, as shown in Fig. 2. It was apparent that both electron-donating and electron-withdrawing groups at the *para*- and *meta*-positions of the iodobenzenes were compatible, and the corresponding

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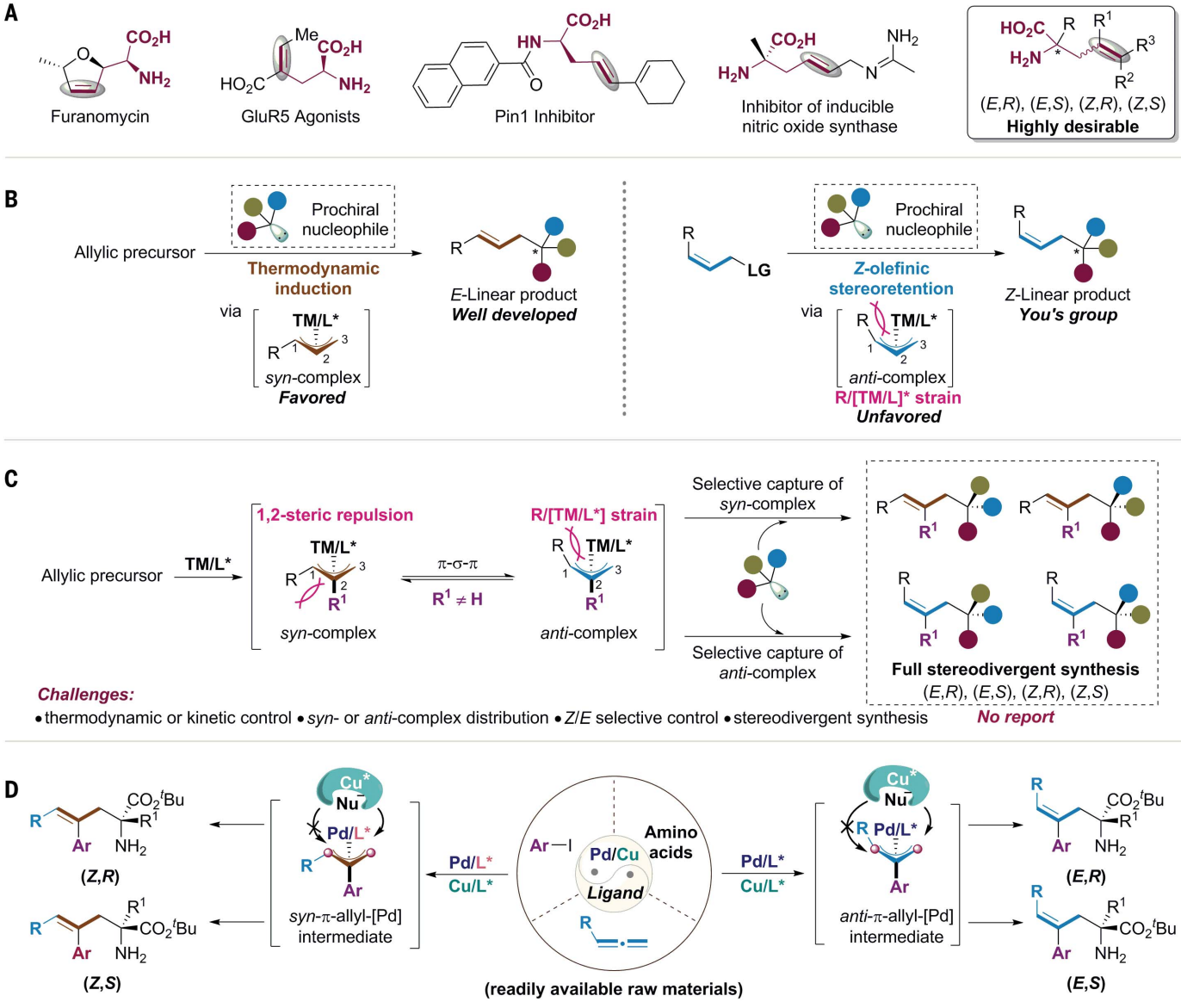



**Fig. 1. Non-natural chiral α-quaternary amino acids bearing trisubstituted alkenes and catalytic control of enantioselectivity and *Z*/*E* selectivity.** (**A**) Representative examples of biologically active α-amino acids bearing alkene moieties. (**B**) The synthesis of enantiopure *E*- or *Z*-disubstituted olefinic compounds bearing chiral centers through 1-monosubstituted π-allyl metal intermediates. (**C**) The challenge for the stereodivergent synthesis of all four stereoisomers of enantiopure *E*- or *Z*-trisubstituted alkenes through 1,2-disubstituted π-allyl metal intermediates. (**D**) The three-component assembly process for the stereodivergent synthesis of enantiopure non-natural α-quaternary amino acids bearing trisubstituted alkene moieties through Pd/Cu dual catalysis. R, alkyl or aryl group; TM, transition metal; L*, chiral ligand; LG, leaving group.

products **4b** to **4p** were obtained in good to high yields, with excellent *Z* and *E* selectivities (92:8 to 98:2 *Z*/*E* and 96:4 to >99:1 *E*/*Z*) and enantioselectivities (95 to >99% e.e.). *Ortho*-substituted iodobenzenes (**4q** and **4r**) were found to be incompatible with this transformation because of steric hindrance. Aryl iodides of pharmaceutically important (hetero)cycles, such as benzodioxole, benzocyclopentanone, phthalide, indole, indazole, naphthalene, pyridine, pyrazole, and thiophene, displayed excellent compatibility, providing products **4s** to **4aa** in good to excellent yields with good to high *Z* and *E* selectivities and enantioselectivities. Iodobenzene tethered to bioactive molecules—including naproxen, ibuprofen, borneol, menthol, galactopyranose, ribofuranosylpurine, citronellol, gemfibrozil, δ-tocopherol, and steroids such as dehydroepiandrosterone, cholesterol, and estrone—all reacted smoothly in this dual-catalytic system, giving the target products **5a** to **5l** in high yields with excellent *Z* or *E* selectivities and diastereoselectivities [>20:1 diastereomeric ratio (d.r.)].

Next, the scope of the allenes **2** was also explored for this stereodivergent process. A series of either electron-donating or electron-withdrawing substituents on the aryl ring of the aromatic allenes at the *para*- and *meta*-positions (**2b** to **2i**) performed well, and the corresponding *Z*- and *E*-trisubstituted alkenes **6b** to **6i** were obtained in high yields (57 to 98%) with excellent *Z* and *E* selectivities (92:8 to 98:2 *Z*/*E* and 92:8 to >99:1 *E*/*Z*) and enantioselectivities (>97% e.e.). The aromatic allene bearing an *ortho*-Cl-substituent on the aryl ring afforded the *E*-selective product (*E*)-**6j** in good yields with excellent enantioselectivities and nearly complete *E* selectivity,

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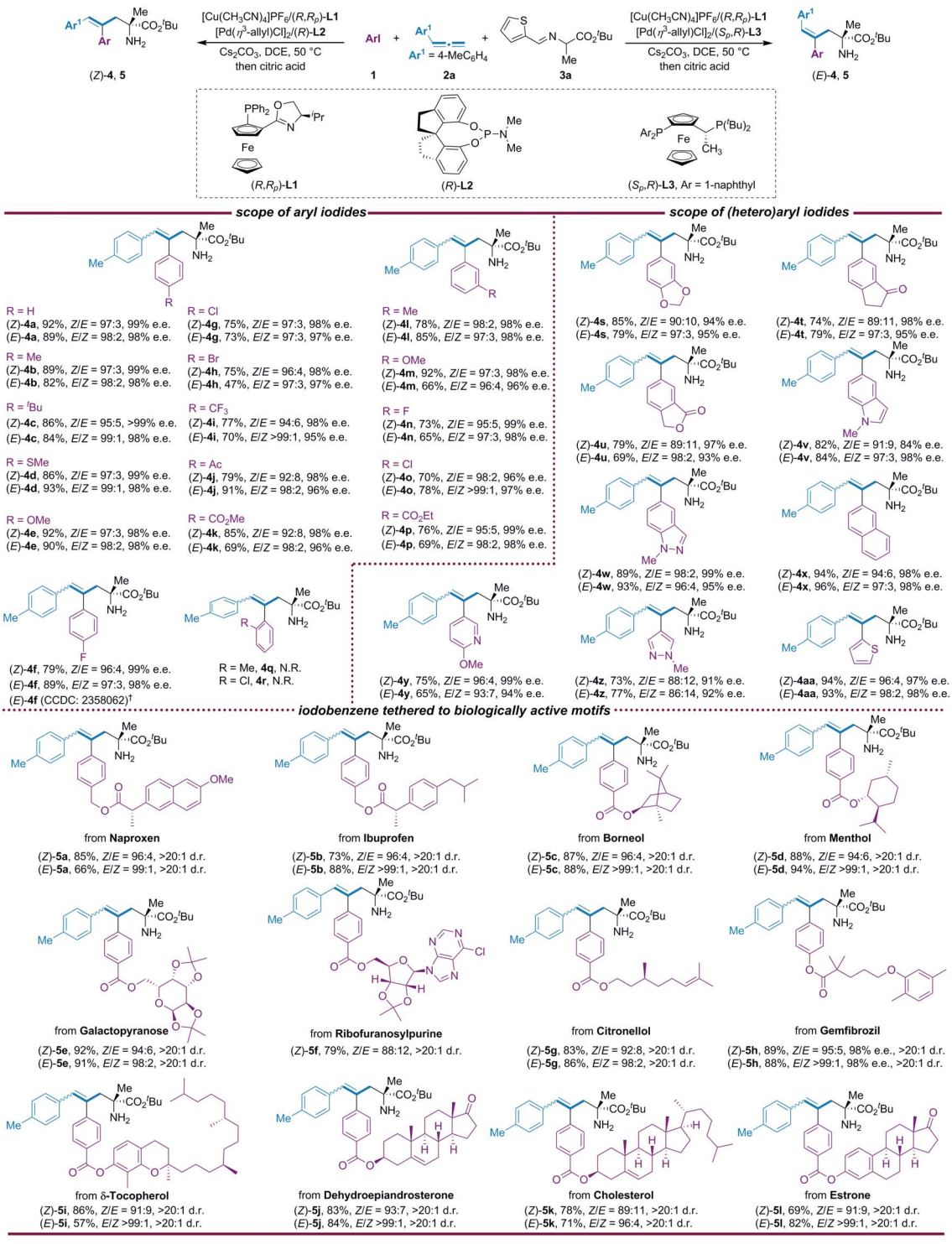



**Fig. 2. Substrate scope of aryl iodides.** Reactions were conducted on a 0.2-mmol scale. Yield refers to isolated, purified mixtures of *E*- and *Z*-isomers. *E*/*Z* ratio was determined by $^1$H NMR spectroscopy. Enantiomeric excess (e.e.) value was determined by high-performance liquid chromatography analysis. †The configuration of (*E*)-**4f** was confirmed by x-ray diffraction analysis of its derivative, (*E*)-**4f**-urea (fig. S16).

whereas poor *Z* selectivity was observed under the *Z*-selective conditions. (Hetero)aryl-substituted allenes also worked well, yielding the desired products bearing naphthalene (**6k** and **6l**), thiophene (**6m** and **6n**), pyrazol (**6o**), pyridine (**6p**), imidazole (**6q**), quinoline (**6r**), and indole (**6s**) groups in moderate to high yields with excellent *Z* and *E* selectivities and enantioselectivities. Additionally, a cyclohexenyl substituted allene was compatible with this dual-catalytic system, giving the desired products (*Z*)-**6t** and (*E*)-**6t** with excellent stereoselectivities. This method provides an avenue for the rapid synthesis of conjugated dienes bearing α-quaternary amino acids. Alkyl-substituted allenes were effective substrates for this transformation as well, affording the desired products **6u** to **6y** smoothly. The β-H elimination by-products were not observed.

Subsequently, an array of α-substituted aldimine esters **3** derived from natural and non-natural amino acids were evaluated. A series of amino acid derivatives containing *Z*- and *E*-trisubstituted alkenes have been prepared, as described in Fig. 3. The racemic aldimine ester substrates bearing alkyl and benzyl groups gave the corresponding products **7b** to **7e** in high yields with satisfactory *Z* and *E* selectivities

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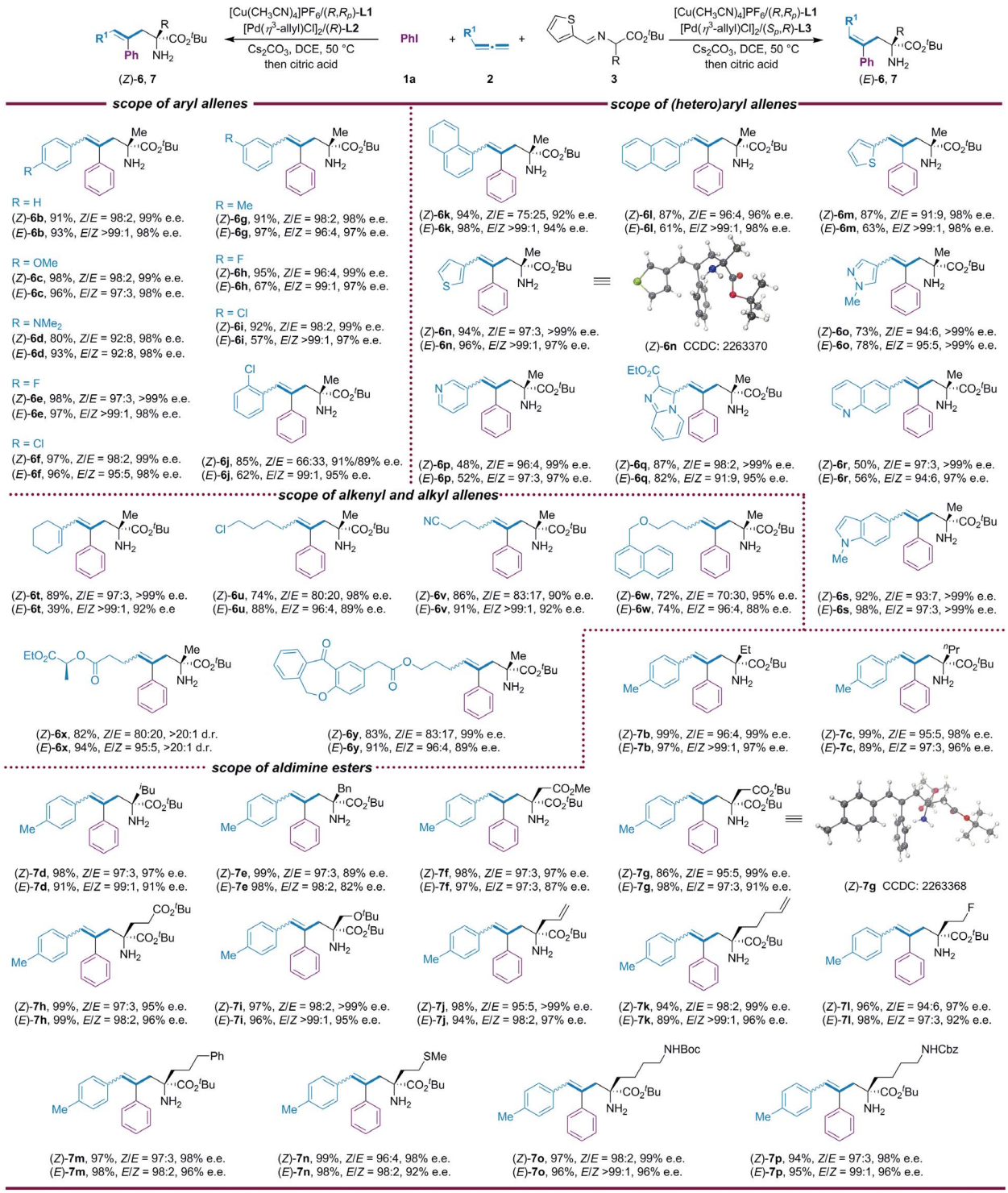



**Fig. 3. Substrate scope of allenes and aldimine esters.** Reactions were conducted on a 0.2-mmol scale. Yield refers to isolated, purified mixtures of *E*- and *Z*-isomers. *E*/*Z* ratio was determined by $^1$H NMR spectroscopy. E.e. value was determined by HPLC analysis.

and enantioselectivities. Diverse synthetically attractive functional groups such as esters (**3f**, **3g**, and **3h**), *tert*-butyl ether (**3i**), allyl (**3j**), alkenyl (**3k**), fluoroalkane (**3l**), phenyl (**3m**), thioether (**3n**), and amides (**3o** and **3p**) were compatible in delivering the corresponding products **7f** to **7p** in high yields (86 to 99%) with excellent *Z* and *E* selectivities (94:6 to 98:2 *Z*/*E* and 97:3 to >99:1 *E*/*Z*) and enantioselectivities (87% to >99% e.e.).

## Full stereodivergent synthesis and synthetic applications

To further evaluate the full stereodivergence of the established protocol, the three-component assembly reactions of **1a**, **2a**, and **3a** were carried out in an enantiodivergent and *Z*/*E*-stereodivergent manner under the optimal reaction conditions. All four stereoisomers of the enantioenriched non-natural α-quaternary amino acids bearing trisubstituted alkene moieties could be smoothly prepared in high yields with excellent *Z* and *E* selectivities and enantioselectivities using the same starting materials

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by simply changing the configuration combinations of the chiral ligands (Fig. 4A). Moreover, to demonstrate the synthetic utility of this dual protocol, gram-scale reactions were performed with no obvious loss in efficiency and stereoselectivities. The products (*Z*,*R*)-**4a** and (*E*,*R*)-**4a** were obtained in 89% yield, 97:3 *Z*/*E* selectivity, 99% e.e. and 88% yield, >99:1 *E*/*Z* selectivity, and 97% e.e., respectively (Fig. 4B). The C=C bonds and amino acid moieties of the products served as versatile functional groups, which could be readily applied for the synthesis of a range of high–value added compounds (Fig. 4C). At first, several transformations with regards to the

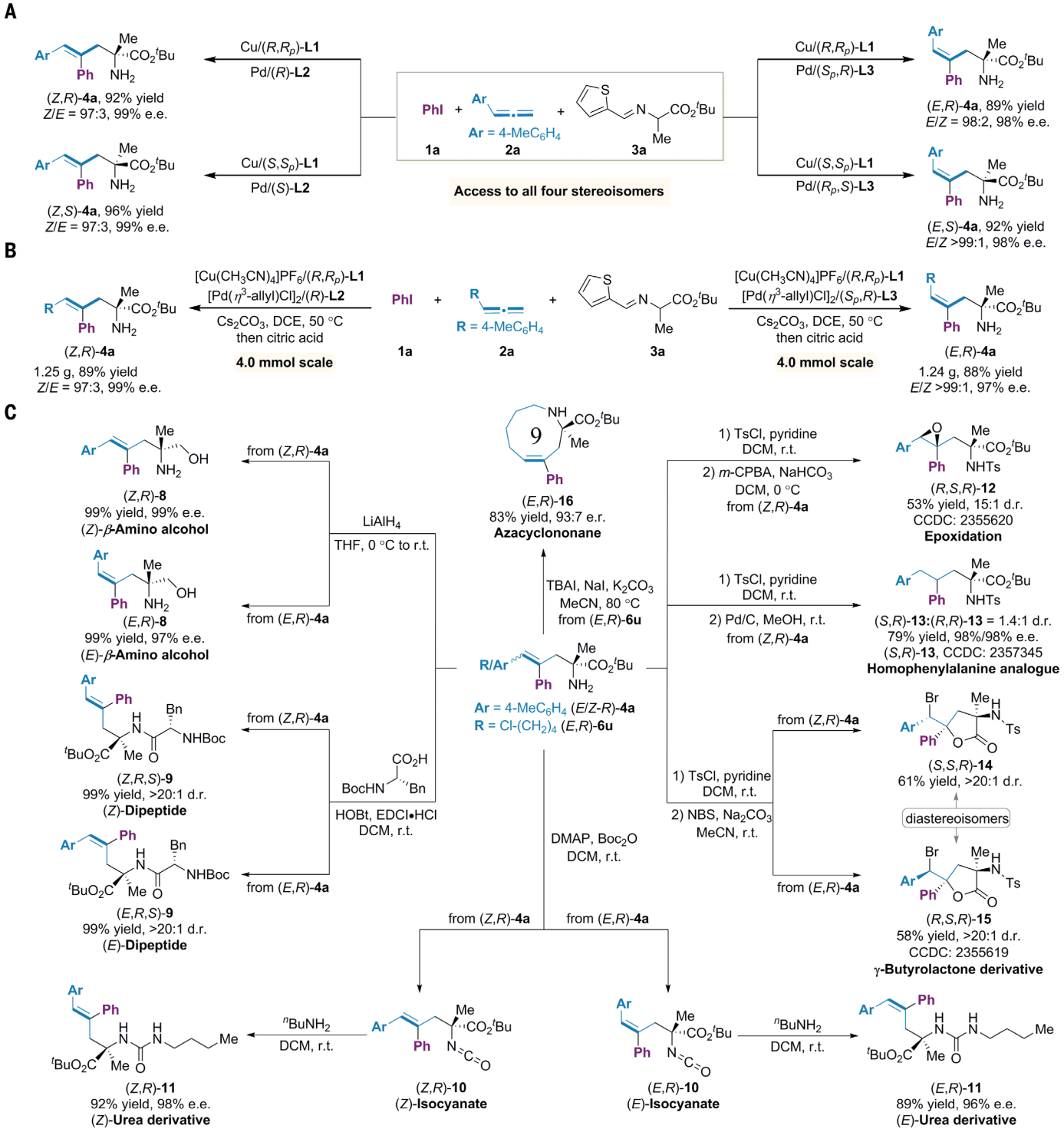



**Fig. 4. Full stereodivergent synthesis of all four stereoisomers, scale-up experiments, and synthetic applications.** (**A**) Synthesis of all four stereoisomers. (**B**) Scale-up reactions for both the *Z*- and *E*-isomers. (**C**) Synthetic transformations of enantioenriched non-natural α-quaternary amino acids bearing trisubstituted alkene moieties.

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amino ester moieties of **4a** were conducted, and β-amino alcohols **8**, dipeptides **9**, isocyanates **10**, and urea derivatives **11** bearing *Z*- or *E*-trisubstituted alkene groups were successfully produced. The epoxidation and hydrogenation of (*Z*,*R*)-**4a** occurred smoothly to generate the corresponding products **12** and **13** in 53 and 79% yields, respectively. Treatment of (*Z*,*R*)-**4a** or (*E*,*R*)-**4a** with NBS/$Na_2CO_3$ in acetonitrile resulted in the formation of the two diastereoisomers of γ-butyrolactone derivatives **14** and **15** bearing multiple stereocenters in good yields with excellent diastereoselectivities. Lastly, the medium-sized ring compound **16** containing an *E*-olefin could be easily synthesized in 83% yield with 93:7 enantiomeric ratio (e.r.) under alkaline conditions through an intramolecular cyclization reaction of (*E*,*R*)-**6u**.

### Mechanistic investigations

To gain insights into the mechanism of this ligand-controlled stereodivergent process, several mechanistic experiments were conducted. At first, the reactions were monitored over time under the *E*- and *Z*-selective conditions, respectively (figs. S2 and S3). There was no change in the *E*/*Z* ratios and e.e. values of the products (*E*,*R*)-**6b** and (*Z*,*R*)-**6b** as the reaction proceeded. These results suggest that the *E*/*Z* isomerization of trisubstituted alkenes does not occur, thus further supporting the ligand control for stereoselectivity in our dual-catalytic system (*52*, *53*). To understand the different behaviors of **L2**- and **L3**-ligated π-allyl-Pd intermediates, additional experiments were performed. In general, the characterization of *anti*- or *syn*-π-allyl-Pd complexes could provide valuable information for further understanding the origin of the *E* or *Z* selectivity (*39*, *54*). Fortunately, the *anti*-π-allyl-Pd/**L3** intermediate **17** was successfully obtained as a single isomer, which was determined by x-ray diffraction analysis (Fig. 5A). No isomerization of this *anti*-π-allyl-Pd/**L3** complex was observed at 50°C, even over 36 hours (fig. S4). Then, the stoichiometric complex **17** was reacted with the aldimine ester **3a** using the Cu/**L1** catalyst, affording the product **4a** with nearly complete *E* selectivity (Fig. 5A). Subsequently, pure (*Z*)- and (*E*)-2-aryl-substituted cinnamyl carbonates **18** and **19** were examined under standard *E*-selective conditions, which could, in principle, initially generate the *anti*- and *syn*-π-allyl-Pd/**L3** intermediates through the ionization process, respectively (Fig. 5B). Regardless of whether (*Z*)-**18**/**19** or (*E*)-**18**/**19** were used, the *E*-selective products (*E*)-**6b**/**20** were predominant. Moreover, both (*Z*)- and (*E*)-**19** were reacted with a stoichiometric amount of the $Pd^0$/**L3** catalyst, and the reactions were monitored by $^{31}P$ and $^{19}F$ nuclear magnetic resonance (NMR) spectroscopy (figs. S5 and S6). It was observed that (*Z*)-**19** preferentially generated the *anti*-π-allyl-Pd/**L3** intermediate, which increased in concentration over time. By contrast, (*E*)-**19** initially formed the *syn*-π-allyl-Pd/**L3** intermediate, which readily converted into the *anti*-π-allyl-Pd/**L3** intermediate. These results indicate that the formation of the *anti*-π-allyl-Pd/**L3** is thermodynamically favored.

To gain a deeper understanding of the origin of the observed *E* selectivity in the current Pd/**L3** catalytic system, kinetic experiments and density functional theory (DFT) calculations were conducted. First, the migratory insertion process of the allene moiety into the aryl-Pd bond was examined through $^{31}P$ NMR monitoring experiments involving the reaction of the allene with stoichiometric aryl-$Pd^{II}$ species (figs. S7 and S8). The results showed that the *anti*-π-allyl-Pd/**L3** species was preferentially formed without isomerization, indicating that this species also acts as a kinetic intermediate. Additionally, a kinetic study on the concentration of the copper catalyst revealed a zero-order rate dependence, suggesting that the nucleophilic attack process is not the rate-determining step (fig. S15). These experimental results imply that the migratory insertion is the *E*-selective stereocontrol step. Subsequently, DFT computations for all eight complete reaction pathways were investigated (fig. S24). The free-energy profiles of two favored pathways for the formation of *E*- and *Z*-selective products are shown in Fig. 5C. Both pathways begin with the $Pd^0$/**L3** complex **IM-(*E*)-1**, followed by oxidative addition via **TS-(*E*)-1a**, forming the aryl-Pd/**L3**-I species **IM-(*E*)-2a**. A ligand-exchange process between **IM-(*E*)-2a** and allene **2b** then occurs, affording **IM-(*E*)-3ab** and **IM-(*E*)-3ac**. Subsequent migratory insertion in **IM-(*E*)-3ab** and **IM-(*E*)-3ac** generates **IM-(*anti*)-4ab** and **IM-(*syn*)-4ac**, respectively. The calculations reveal that the energy barrier for the formation of **IM-(*anti*)-4ab** is obviously lower than that for the formation of the **IM-(*syn*)-4ac** by 10.48 kcal/mol [**TS-(*E*)-3ab** versus **TS-(*E*)-3ac**], owing to the considerable steric hindrance between the allene and ligand moieties in **TS-(*E*)-3ac** (fig. S28). Lastly, the copper-chelated nucleophile **Nu** attacks the cationic intermediate **IM-(*anti*)-4ab** via **TS-(*anti*)-4ab**, leading to the major product (*E*)-**6b**. The DFT data are in good agreement with the experimental results. The above experimental and computational results suggest that the migratory insertion step is the *E*-selective stereocontrol step, which preferentially generates the thermodynamically and kinetically favored *anti*-π-allyl-Pd/**L3** intermediate **IM-(*anti*)-4ab**. The cationic intermediate **IM-(*anti*)-4ab** can be directly captured by the nucleophile to deliver the *E*-product.

Unfortunately, the *syn*-π-allyl-Pd/**L2** intermediate could not be isolated but was detected by high-resolution mass spectrometry and $^{31}P$ NMR spectroscopy (figs. S9 and S10) (*55*). To determine the origin of the observed *Z* selectivity in the Pd/**L2** catalytic system, detailed mechanistic studies were performed. First, control experiments with pure (*E*)- and (*Z*)-2-Ph-substituted cinnamyl carbonates **18** were conducted, affording the same *Z*-products with essentially the same *Z*/*E* ratios (Fig. 5D). Moreover, kinetic studies on the reactions with allylic carbonate **18** and allene **2a** showed that both *syn*- and *anti*-intermediates were generated for the two substrates and then rapidly interconverted in the Pd/**L2** catalytic system, indicating that the nucleophilic attack is the stereo-determining step (figs. S10 and S11). Subsequently, DFT calculations for all eight complete reaction pathways were investigated (fig. S25). The free-energy profiles of two favored pathways for the formation of (*E*)- and (*Z*)-**6b** are shown in Fig. 5E. Both pathways start from the $Pd^0$/**L2** intermediate **IM-(*Z*)-1**, which undergoes an oxidative addition through **TS-(*Z*)-1** to form the aryl-$Pd^{II}$/**L2**-I intermediate **IM-(*Z*)-2**. The subsequent ligand exchange between **IM-(*Z*)-2** and allene **2b** generates **IM-(*Z*)-3c** and **IM-(*Z*)-3h** (fig. S27). Next, the migratory insertion produces **IM-(*anti*)-4c** and **IM-(*syn*)-4h** via **TS-(*Z*)-3c** and **TS-(*Z*)-3h**, respectively. DFT studies suggest that the interconversion between π-allyl-Pd/**L2** intermediate **IM-(*anti*)-4c** and **IM-(*syn*)-4h** occurs through an $\eta^3$-$\eta^1$-$\eta^3$ process facilitated by the iodide ion (Fig. 5E, bottom). Lastly, the copper-chelated nucleophile **Nu** selectively attacks the neutral π-allyl-Pd/**L2** intermediate **IM-(*anti*)-4c** or **IM-(*syn*)-4h** through **TS-(*anti*)-4c** or **TS-(*syn*)-4h**, leading to the product **6b**. The energy barrier for the formation of (*Z*)-**6b** is 1.51 kcal/mol lower than that for the formation of (*E*)-**6b** [**TS-(*anti*)-4c** versus **TS-(*syn*)-4h**], which mostly aligns with experimental findings. In addition, energy decomposition analysis of **TS-(*anti*)-4c** and **TS-(*syn*)-4h** reveals that the stronger attractive energies and lower distortion energies present in **TS-(*syn*)-4h** contribute to the favored *Z*-product (fig. S33). All these results indicate that selective nucleophilic attack to the *syn*-π-allyl-Pd/**L2** intermediate is the *Z*-selective stereocontrol step, and a specific pathway for the *Z* selectivity is proposed as follows: Both *anti*- and *syn*-π-allyl-Pd/**L2** intermediates could be formed by means of the migratory insertion of the allene moiety into the aryl-Pd bond. The *anti*-π-allyl-Pd/**L2** intermediate is then converted into the *syn*-π-allyl-Pd/**L2** intermediate through an $\eta^3$-$\eta^1$-$\eta^3$ process, which is captured by the nucleophile to deliver the *Z*-product.

In addition, control experiments were conducted to elucidate the roles of the chiral palladium and copper catalysts in the observed *Z*/*E* selectivity and enantioselectivity (table S3). The results demonstrate that the chiral copper catalyst predominantly controls the enantioselectivity for both Pd/**L2** and Pd/**L3**

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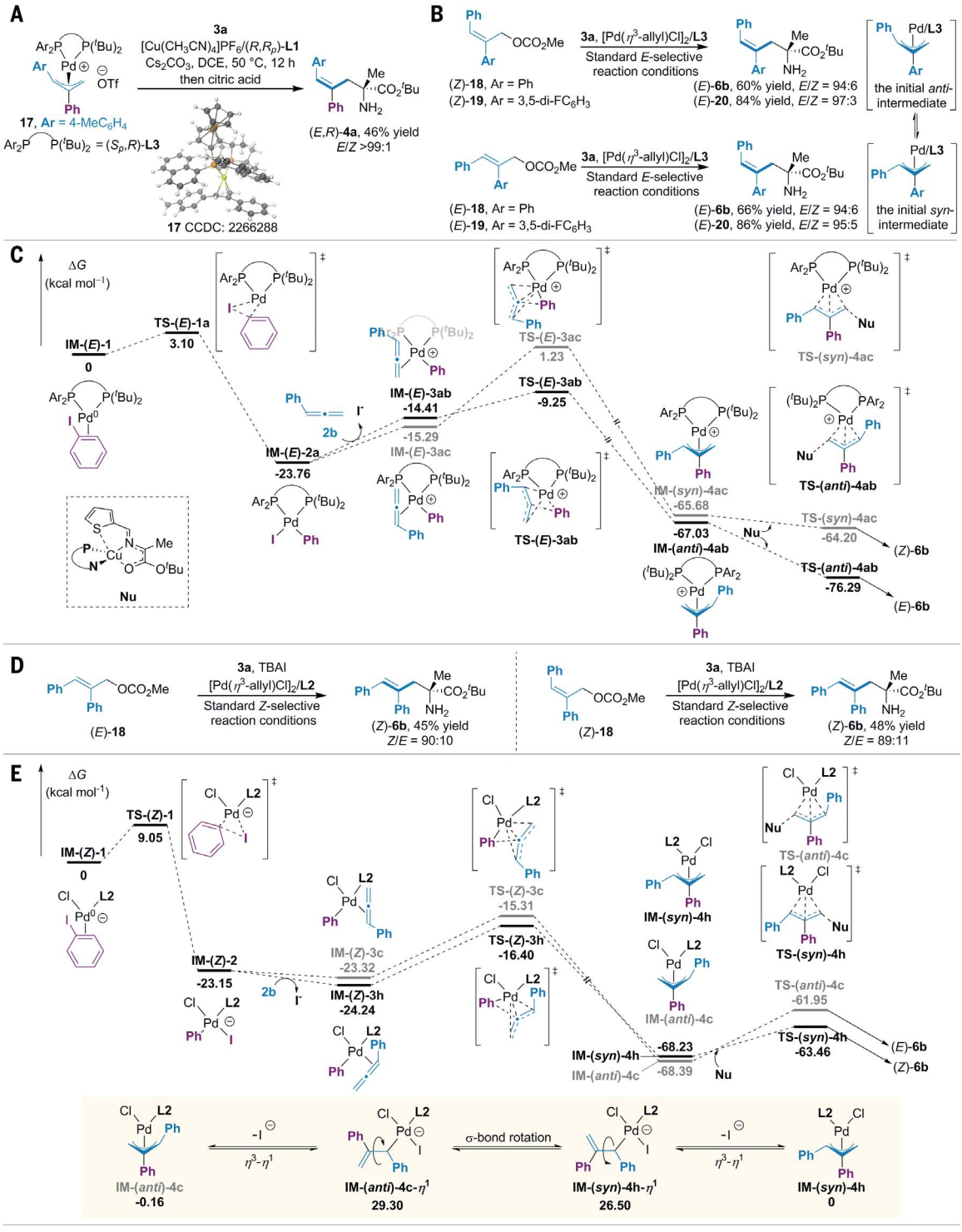



**Fig. 5. Mechanistic studies.** (**A**) Evaluation of the complex **17** and its single-crystal structure. (**B**) Control experiments with (*E*)- and (*Z*)-**18**/**19** under the *E*-selective conditions. (**C**) DFT calculations of the Pd/**L3** catalytic system. (**D**) Control experiments with (*E*)- and (*Z*)-**18** under the *Z*-selective conditions. (**E**) DFT calculations of the Pd/**L2** catalytic system.

catalytic systems (table S3, entries 5 to 7). Moreover, an obvious match/mismatch effect was observed when replacing (*R*)-**L2** with (*S*)-**L2** or (*S*$_p$,*R*)-**L3** with (*R*$_p$,*S*)-**L3** (table S3, entries 8 and 9). Furthermore, control experiments with different copper ligands in the Pd/**L3** catalytic system show a limited effect of the ligand on the *E* selectivity, implying that the migratory insertion step dominates the *E*-selectivity control (table S3, entries 7 and 9). By contrast, the copper ligand has a significant influence on the *Z* selectivity in the Pd/**L2** catalytic system, suggesting that the nucleophilic attack step plays a crucial role in governing the *Z* selectivity (table S3, entries 6 and 8). These results are qualitatively consistent with the above computational and experimental results for the two catalytic systems (Fig. 5).

The success of the comprehensive control of alkene configuration and point chirality has been realized through synergistic dual chiral metal catalysis. This study not only addresses a long-standing challenge in π-allyl metal chemistry for stereocontrol over both *Z* and *E* configurations of C=C bonds but also provides a potentially general strategy for the stereodivergent synthesis of enantiopure molecules

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bearing *Z*- and *E*-alkene moieties. Furthermore, the exploration of all stereoisomers [(*Z*,*R*), (*Z*,*S*), (*E*,*R*), (*E*,*S*)] of enantioenriched non-natural α-quaternary amino acids, featuring trisubstituted alkene moieties, opens up more possibilities for drug design as well as protein and peptide modifications.

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## SUPPLEMENTARY MATERIALS

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# Stereodivergent access to non-natural #-amino acids via enantio- and *Z*/*E*-selective catalysis

Panpan Li, En Zheng, Guanlin Li, Yicong Luo, Xiaohong Huo, Shengming Ma, and Wanbin Zhang

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**Editor's summary**

Adaptable control over the disposition of groups at each end of a carbon-carbon double bond remains a challenge in chemical synthesis. In the *E* configuration, the largest groups are diagonally opposed, whereas in the *Z* configuration, they're on the same side. Li *et al*. paired up copper and palladium catalysts to deliver either *E*- or *Z*-configured double bonds in unnatural amino acids, depending on the ligand choice. The reaction is also highly enantioselective at the quaternary alpha carbon stereocenter. —Jake S. Yeston

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